

Alexander Cristhoper Trujillo Ochoa

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Experience

Leonardo S.p.A.

HPC Specialist

Genova, Italy

Apr 2023 — Present

- Developed and parallelized new CFD functionalities in C++, FORTRAN, and Python for the helicopter division's aerodynamic simulation models
- Established CI/CD and build pipelines for legacy codebases, enabling deployment on modern HPC clusters with automated testing and coverage
- Benchmarked performance across 3+ CPU architectures, identifying optimization targets that enhanced numerical precision and execution efficiency
- Modernized legacy FORTRAN codes to leverage MPI and OpenMP parallelism, reducing computation time for production CFD workflows

DIII-D & UC San Diego

Research Intern

San Diego, USA

Jun — Oct 2019

- Optimized waveguide transmission line simulations (~100 GHz) for the Electron Cyclotron Heating system, improving plasma confinement model accuracy
- Managed the pyDIII-D framework for plasma physics experiment validation across the ECH research group
- Awarded **Outstanding Student Research Experience** by UCSD (May 2019)

Research

Peruvian Commission for Aerospace Research

Researcher

Lima, Peru

Dec 2018 — Jun 2019

- Analyzed elementary particle data in cosmic ray air showers using PyRoot (Python & C++) to assess gamma ray detection viability
- Optimized GEANT4 simulations of Cherenkov tanks at Huancayo's magnetic observatory, improving detector modeling accuracy

UNAM Institute of Physics

Research Intern

Mexico City, Mexico

Oct 2017 — Jun 2018

- Simulated detectors for material characterization by backscattering using GEANT4, based on the Institute's mass spectrometer
- Developed C++ simulation code and authored technical documentation for the research group

Education

ICTP — SISSA

Master in High Performance Computing

Trieste, Italy

Sep 2021 — Jan 2023

Universidad Nacional de Ingeniería

B.Sc. Physics — First in Class

Lima, Peru

Aug 2013 — Dec 2018

MITx

Micromasters in Statistics and Data Science

Online

Apr 2020 — Jul 2021

- Completed training in AI development and data-driven prediction using Python, PyTorch, and TensorFlow

Skills & Additional

HPC:	MPI, OpenMP, CUDA, SYCL, Ansible, Slurm, Bright, Linux
Programming:	C, C++, Python, FORTRAN, Shell Scripting
Frameworks:	GEANT4, Deal.II, PetSc, MATLAB, COMSOL, ANSYS
AI/ML:	PyTorch, TensorFlow, Google AI Platform, pandas, R
DevOps:	Git, GitLab/GitHub, CI/CD Pipelines, Testing, Coverage
Languages:	Spanish (Native), English (TOEFL 102, Advanced), Italian (Advanced)